INCH-POUND

MIL-C-71155 (AR)
01 November 1993

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M987
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers requirements and quality assurance provisions for the loading, assembling and packing for the Cartridge, Ignition M987 for use on the Cartridge, 81MM; Practice, Short Range, M766.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414	-	Sampling Procedures and Tables for Inspections by Variables for Percent Defective

(Unless otherwise indicated, copies of federal and Military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect of the date of the solicitation.

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12953580 - Cartridge, Ignition, M987

INSPECTION EQUIPMENT DRAWINGS

9278622 - Envelope Drawing for Leak Testing of Ignition Cartridges

(Copies of other Government documents, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity or as directed by the contracting officer).

CODE OF FEDERAL REGULATIONS

49 CFR - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite 49 CFR (latest revision)).

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The ignition cartridge assembly shall comply with all requirements specified on drawing 12953580 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Total volatiles content of bullseye powder. The total volatiles content of the bullseye powder used in loading the ignition cartridge at the time of loading, shall not exceed 2.0 percent. Testing shall be as specified 4.5.1.

3.4 Sensitivity.

3.4.1 Nonfunctioning. The primer shall not function under an applied energy level of 8.85 inch ounces (min) when tested as specified in 4.5.2.1.

3.4.2 Functioning. The primer shall function under an applied energy level of 82.6 inch ounces (max.) when tested as specified in 4.5.2.2.

3.4.3 Safety drop test. The primer shall not function as a result of the 5 foot cartridge drop as specified in 4.5.2.3

3.5 Leak test. The loaded cartridges shall satisfactorily pass the leak test specified in 4.5.3.

3.6 Velocity, functioning and security (proving ground test). The ignition cartridges shall comply with the following requirements (See 4.5.5).

3.6.1 Misfires. The ignition cartridges shall function on the first attempt, without any audible hangfire.

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3.6.2 Metal parts security. There shall be no evidence of rupture or metal parts separation attributable to ignition cartridge functioning.

3.6.3 Velocity and functioning (proving ground). The cartridge assemblies shall function and shall impart a charge 3 muzzle velocity of 75.00 ± 2.00 mps at +70 ± 3 degrees F. In addition, the velocity standard deviation of the lot shall be not greater than 4 fps.

3.7 Propellant charge establishment & uniformity.

3.7.1 Propellant charge establishment. The propellant slope (see 4.5.6) shall be determined for each propellant lot which will yield a service velocity of 75.00 mps at charge 3 at a temperature of +70 ± 3 degrees F. This requirement shall be met for first article and each propellant lot.

3.7.2 Propellant charge uniformity. The propellant loading weight shall be determined for each propellant lot to meet the required velocity requirements (see 3.6.3 and 4.5.6.2). This requirement shall be met for first article and each propellant lot.

3.7.3 Propellant charge uniformity (two year lapse). The propellant lots shall achieve a mean velocity of 75.00 mps ± 2.00 mps at charge 3 at a temperature of +70 ± 3 degrees F (See 4.5.6). This test shall be required when a lapse of two years or more occurs between propellant acceptance and LAP initiation.

3.8 Static pressure. The average internal ignition cartridge pressure plus 4 standard deviations shall not exceed 22000 psi (See 4.5.7). This requirement shall be met for first article and for each propellant lot.

3.9 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with technical provisions herein (See 4.3 and 6.2).

3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The requirements of MIL-A-48078 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the components in the quantities indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

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4.3.3 Rejection. See MIL-A-48078 and the first article velocity test shall be rejected if:

a. Two or more misfires occur.

b. One or more short rounds occur.

c. One or more premature burst occur.

d. One or more metal parts separation occur.

e. Velocity requirements are not met. (70 degrees F conditioned samples only).

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71155 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, Ignition, M987	1 OF 2	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Propellant (a) (dwg. 12944156)			
	Charge establishment (b)	9	3.7.1	4.5.6.1
	Charge uniformity (b)	30	3.7.2	4.5.6.2
	Static pressure (b)	25	3.8	4.5.7

NOTES:

(a) The First article requirements shall be performed in two stages. The establishment and uniformity phase shall be performed, the loading charge determined, and then the remaining first article inspections & tests performed and requirements met.

(b) This test shall be performed and requirements met for each propellant lot in addition to first article.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71155

PARAGRAPH	TITLE Cartridge, Ignition, M987	SHEET 2 OF 2	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Shell, Shotgun, Primed, 20 Gauge Assy. (Dwg. 12953575)			
	Examination for defects	20	3.2	4.4.2.1
	Sensitivity test	20 (a)	3.4.1/ 3.4.2	4.5.2
	Cartridge, Ignition, M986, (prior to addition of powder) (Dwg. 12953580)			
	Examination for defects	20	3.2	4.4.2.2
	Total volatile content of powder	5 Ctgs	3.3	4.5.1
	Cartridge, Ignition, M987 Assembly (Dwg. 12953580			
	Examination for defects	125	3.2	4.4.2.3
	Sensitivity safety drop test	25 (b)	3.4.3	4.5.2.3
	Leak test	50 (b)	3.5	4.5.3
	Velocity, functioning and security	50 (b)	3.6	4.5.3

NOTES:
(a) The same primed shotgun shells from the examination for defects above shall be used.
(b) The same shotgun shells from the examination for defects above should be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of ignition cartridges shall contain:

a. Metal parts of one lot interfix from one manufacturer.

b. Percussion elements from not more than one lot interfix from one manufacturer.

c. Propellant from not more than one lot from one manufacturer.

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4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Where cited herein, attributes sampling inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs:

TABLE II - Attributes sampling inspection

LOT SIZE	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150000 to 500000	1250	500	200	125	50	13
500000 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71155 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Shell, Shotgun, Primed, 20 Gauge Assembly	SHEET 1 OF 1	12953575
			NEXT HIGHER ASSEMBLY 12953580

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Sensitivity non-function	(a)	3.4.1	Test/4.5.2
Major				
101	Percussion primer missing or loose	III	3.2	Visual/Manual
102	Percussion primer inverted	III	3.2	Visual
103	Depth to primer from face	III	3.2	Gage
104	Sensitivity function	(a)	3.4.2	Test/4.5.2
Minor				
201	Poor workmanship	V	3.10	Visual

NOTES:
(a) Sample per 4.4.3.2.1

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71155 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cartridge, Ignition, M987 Prior to Addition of Powder	1 OF 1	12953580
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Flash tube improperly placed, damaged, or not sealed closed	III	3.2	Visual
102	Inner tube improperly placed	III	3.2	Visual
103	Total volatile content of powder	(a)	3.3	Test
Minor				
201	Poor workmanship	V	3.10	Visual

NOTES:

(a) Sample shall be selected from each eight (8) hour production interval.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Cartridge, Ignition, M987 Assembly	1 OF 2	12953580

NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Weight of propellant (before loading)	100%	4.4.3.1.	Gage
B	Weight of propellant (after loading) more than 3 grains above assessed maximum weight	(a)	3.2	Gage
C	Weight of propellant (after loading) more than 3 grains below assessed minimum weight	(a)	3.2	Gage
D	Sensitivity safety drop test	25 (d)	3.4.3	4.4.3.2.3/4.5.2.3
Major				
01	Hot melt sealant missing, inadequate or excessive	III	3.2	Visual
02	Length, max.	III	3.2	Gage
03	Crimp missing or incomplete	III	3.2	Visual
04	Leak test	100%	3.5	4.5.3
05	Velocity, functioning, and security (proving ground)	(b)	3.6	4.5.5
06	Marking missing or unidentifiable	III	3.2	Visual
07	Diameter, max.	III	3.2	Gage
08	Environmental conditions inadequate	(c)	3.2	Gage

NOTES:

(a) See 4.4.3.1.2, Same sample used for both inspections.

(b) See 4.4.3.3.

(c) Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing.

(d) Test required only to qualify each new lot of primers when they are initially introduced in M987 production.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71155 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Cartridge, Ignition, M987 Assembly	2 OF 2	12953580
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
109	Weight of propellant (after loading) three grains or less above assessed max weight	(a)	3.2	Gage
110	Weight of propellant (after loading) three grains or less below assessed minimum weight	(a)	3.2	Gage
Minor				
201	Poor workmanship	V	3.10	Visual

NOTES:

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4.4.3 Testing. See 4.5

4.4.3.1 Weight of propellant.

4.4.3.1.1. Before loading. (See dwg. 12953580) The propellant charge shall be weighed. Weight limits for weighing should be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective and removed from the lot. (See dwg for tolerance requirement).

4.4.3.1.2 After loading. 315 sample cartridge assemblies shall be randomly selected from each lot for this test. If four or more samples exceed the assessed max weight by 3 or less grains or if any sample exceeds the assessed max weight by more than 3 grains, the lot shall be rejected. If four or more samples are under the assessed minimum weight by 3 grains or less or if any sample is under the assessed minimum weight by more than 3 grains, the lot shall be rejected. In addition, if the total number of defective samples are five or more, the lot shall be rejected. Testing shall be as specified in 4.5.4.

4.4.3.1.3 Automated method. The contractor may elect to perform this operation with automated equipment and mandatory process controls. In such cases, process control plans must be submitted to the Technical Agency for approval of the automated operation.

4.4.3.2 Sensitivity.

4.4.3.2.1 Nonfunctioning. Beginning with the first article and first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 200 primer housing assemblies shall be randomly selected from each lot for this test. If one or more assemblies fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.2.1.1 Nonfunctioning. After three consecutive lots have met the criteria of 4.4.3.2.1, the sampling plan for this test shall be in accordance with Level IV. If a lot is rejected, revert to sampling plan per 4.4.3.2.1. The test shall be performed as specified in 4.5.2.1.

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4.4.3.2.2 First article and initial production functioning test. Beginning with the first article and first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 200 primer housing assemblies shall be randomly selected from each lot for this test. If 4 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 1 but less than 4 defectives are found a second sample of 200 primer housing assemblies shall be tested. If the combined number of defectives in both the first and second sample is 4 or more, the lot shall be rejected. Any assemblies in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

4.4.3.2.2.1 Subsequent production functioning test. After three consecutive lots have met the criteria of 4.4.3.2.2, the sampling plan for this test shall be per Level III and shall be randomly selected from each lot for this test. If 1 or more primers fail to comply with the specified requirement, the lot shall be rejected. Any assemblies in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. If a lot is rejected, revert to sampling plan per 4.4.3.2.2. The test shall be performed as specified in 4.5.2.2.

4.4.3.2.3 Safety drop test. Twenty-five ignition cartridge assemblies consisting of the shot gun shell with primer (assembled without the bulls eye powder in accordance with dwg 12953575) shall be selected for this test. If one or more of the test samples function when tested as specified in 4.5.2.3, the primer lot shall be rejected. This test shall be performed only upon initial use of each primer lot or each primed shotgun shell lot.

4.4.3.3 Velocity, functioning and security

4.4.3.3.1 First article testing and initial production. Beginning with the first article and first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter M and a maximum percent defective of 1.17. The upper and lower specification limits shall be 73.00 and 77.00 meters per second (mps), respectively. In addition, if the velocity standard deviation of the sample exceeds the lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. This test shall be performed as specified in 4.5.5.

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4.4.3.3.2 Production lot testing. After three consecutive lots have met the criteria of 4.4.3.3.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter K, using a maximum percent defective of 1.23. The upper and lower specification limits shall be 75.00 ± 2.00 mps.. In addition, if the velocity standard deviation of the sample exceeds the applicable lot requirement, the lot shall be referred. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. If a lot is rejected revert to sampling plan per 4.4.3.3.1.A. This test shall be performed as specified in 4.5.5.

4.4.3.4 Check test for deterioration of primer. If the total elapsed time between original acceptance of the primer lot and the assembly of that lot into the ignition cartridge exceeds two years, or if the primers have been subjected to adverse conditions for any period of time since previous test, the primer lot shall be subjected to and must satisfactorily pass the sensitivity, safety drop, and velocity tests in the first article immediately before the primer lot is assembled into the ignition cartridges.

4.4.3.5 Static pressure. The propellant lot shall be tested as specified in 4.5.7. Twenty-five ignition cartridges shall be loaded at the assessed charge weight with propellant randomly selected from the propellant lot. If the average ignition cartridge pressure exceeds the specified requirement, the representative lot of bullseye propellant shall be withheld and referred to the engineering agency.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Total volatile content of propellant powder. The total volatile content of the powder shall be determined in accordance with the procedure specified in MIL-STD-286. The samples shall be randomly selected from each eight hours production. Any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, ignition cartridges loaded with the nonconforming powder shall be rejected.

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4.5.2 Sensitivity.

4.5.2.1 Nonfunctioning. The shotgun shell with primer assembly shall be assembled to a suitable fixture. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A steel ball shall be dropped onto the firing pin producing 8.85 inch-ounce of energy (min). The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer housing assembly which functions shall be classed defective. Samples used in this test shall not be returned to the lot.

4.5.2.2 Functioning. The shotgun shell with primer assembly shall be assembled to an appropriate test fixture. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug of the primer. A steel ball shall be dropped on to the firing pin producing 82.6 inch-ounce of energy (max). The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer housing assembly that fails to function shall be classed defective.

4.5.2.3 Safety drop test. The test assemblies shall be assembled into inert-loaded, M766 cartridges. Each cartridge shall be dropped twice from a height of 5 feet on an anvil top surface. For the first drop, the cartridge shall be in the base (fin) down orientation. For the second drop, the cartridge shall be in the nose (fuze) down orientation. The drop height shall be measured from the lowest point of the cartridge orientation to the top of the anvil surface. Any test assembly which functions as a result of the drop shall be classed defective. All questionable test assemblies shall be downloaded to verify primer functioning or nonfunctioning. All components used for this test shall not be returned to their respective lots. Each cartridge will have the inert fuze and fin assembly replaced after each double drop.

4.5.3 Leak test. The ignition cartridges shall be tested 100 percent in accordance with drawing 9278622, after a two (2) hour minimum cure time. Any ignition cartridge which fails to comply with applicable requirement shall be classed defective and removed from the lot.

4.5.4 Weight of propellant. 315 random samples of the ignition cartridge will be taken prior to the crimping operation and all propellant removed and placed on the pan of a precision analytical balance. The weight of propellant from each sample shall be determined to the nearest grain.

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4.5.5 Velocity, functioning and security (proving ground test). The test shall be performed at a Government proving ground in accordance with the applicable USA TECOM Acceptance Test Procedure.

a. The sample ignition cartridges shall be assembled with a standard finned M766 projectile (2.9 ± .2 pounds (lbs)) for which they were manufactured. The complete assembly shall be fired from a M224 ballistic tube at an elevation of forty-five degrees. Half the cartridges shall be fired at charge 0 and half at charge 3.

b. Individual velocities shall be recorded on the firing record for the test rounds. Velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day and time of firing shall also be recorded. In the event of a misfire, the misfired ignition cartridge shall not be disassembled by the proving ground but shall be forwarded to the responsible technical activity. Individual chamber pressures shall be obtained for each round firing.

4.5.5.1 Refiring. If for any reason the proving ground considers that conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

4.5.5.2 Referral criteria. If any test rounds fail velocity standard deviation or average pressure requirements at 21 degrees C or whenever the Proving Ground deems necessary.

4.5.6 Propellant charge establishment and uniformity.

4.5.6.1 Propellant charge establishment.

4.5.6.1.1 Charge slope method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70 degrees F, and an angle of fire of 45 degrees using an M224 Mortar. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.7.1). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.6.2). The difference between the velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

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4.5.6.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can then be used as in 4.5.6.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.5.6.2 Propellant charge uniformity. Thirty rounds shall be assembled with the established charge weight of propellant. Ten rounds each shall be conditioned to 0 degrees F, 70 degrees F and 110 degrees F (as indicated by thermocoupled control round) and fired at an angle of fire of 45 degrees in the M224 Mortar. Firing results at all temperatures shall be corrected to standard cartridge as fired weight. The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results.

4.5.6.2.1 Retest criteria. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Test lot fails velocity standard deviation.

b. The final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.5.6.1.

4.5.6.2.2 Referral criteria. A test lot shall be referred for disposition whenever the proving ground deems it necessary. (See 6.8)

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4.5.7 Static pressure test. The ignition cartridges shall be threaded into the Fin and firmly seated. A 0.5 inch hole shall be drilled in the ignition cartridge tube through one flash hole in the forward row of flash holes in the fin boom, counting from the ignition cartridge end. A pressure transducer, mounted to a collar (see figure 1), shall be attached to the fin boom positioning the collar so that the transducer will be directly over the flash hole drilled in the ignition cartridge tube. The collar shall be positioned carefully to prevent restriction of venting from other flash holes in the boom. The assembly shall be set up as shown in figure 2 and the transducer connected to a recorder. The ignition cartridge shall then be functioned and pressure/time trace recorded.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Exterior packing shall be as specified on the applicable drawing

5.1.3 Level C. Same as Level B.

5.2 Marking.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Marking shall be as specified in 49 CFR and the applicable drawing.

5.2.3 Level C. Same as Level B.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The item covered by this specification is intended for assembly in Cartridge, 60MM, Practice SR, M766.

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6.2 Acquisition requirements.

a. See MIL-A-48078.

b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (See 2.1).

c. Provisions for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. See paragraph 6.2.3 of MIL-A-48078. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Submission of test data. One copy of proving ground velocity test data (i.e, firing record) shall be forwarded to the following: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-B, SMCAR-FSS-DM and SMCAR-FSA-MM, Picatinny Arsenal, NJ 07806-5000.

6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Referral criteria. Proving ground referrals shall be sent to U.S. Army Armament Research, Development and Engineering Center (ARDEC), Picatinny Arsenal, NJ 07806-5000, ATTN: SMCAR-QAT-B.

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6.8 Subject term (key word) listing.

Ammunition, mortar
Charge, propelling
Range, precision
Range, standard deviation
Velocity, muzzle
Weight, gross
Weight, propellant

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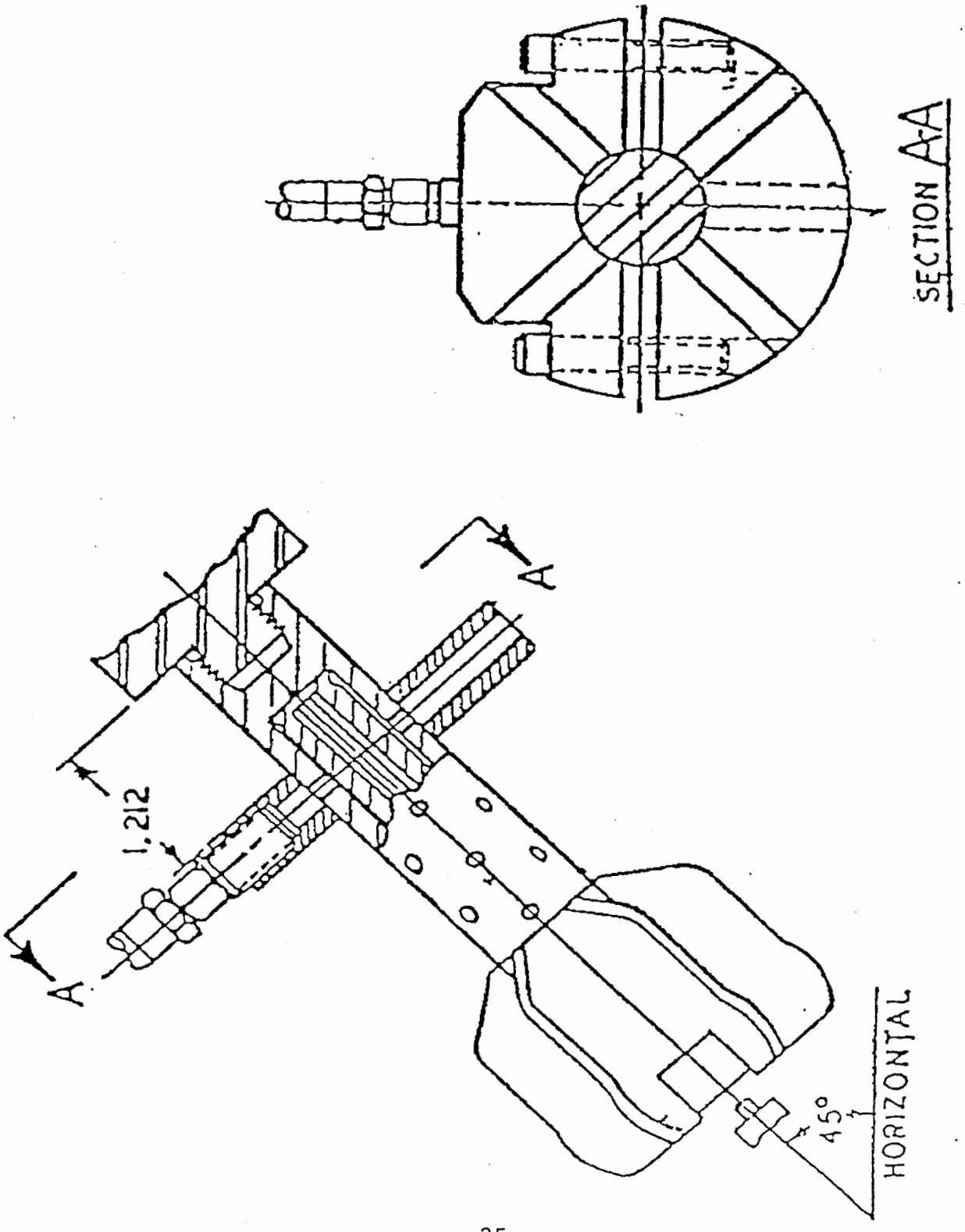


Figure 1. Collar

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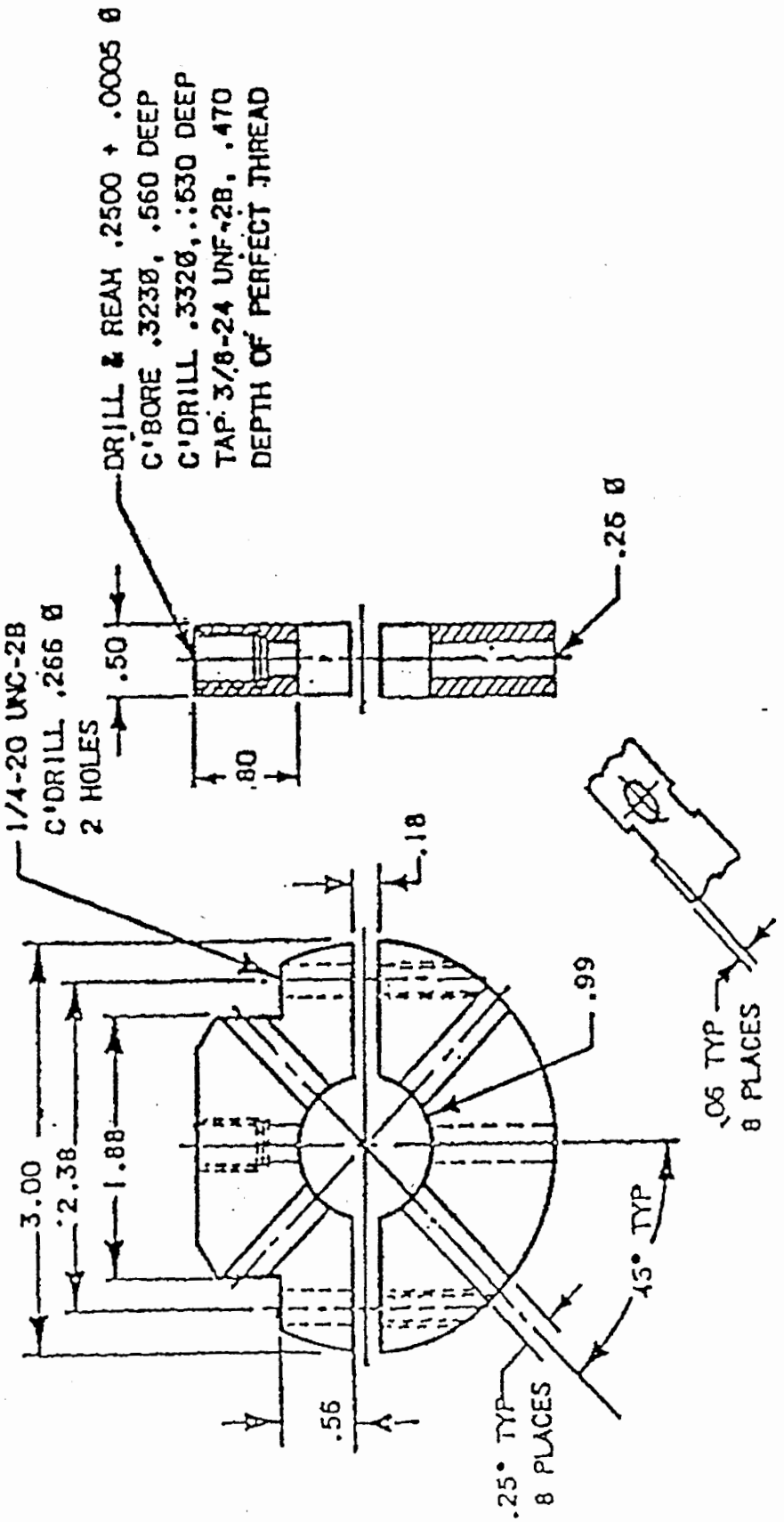


Figure 2. Static Ignition Cartridge Test Setup

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Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AE17)

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